ECE Lab #5: Low Pass and High Pass Filters

Department of Electrical and Computer Engineering
Spring 2025

Overview

The purpose of Lab 5 is to:

- Understand the frequency response characteristics of low pass and high pass filters
- Measure and analyze the magnitude and phase responses of simple RC filters
- Use the M2K Network Analyzer to visualize filter characteristics through Bode plots
- Compare theoretical predictions with actual measurements
- Apply MATLAB to analyze and visualize filter behavior

1 Prelab Assignment

Review material in **Scopy Network Analyzer**

1.1 Filter Fundamentals and Transfer Functions

Review Chapter 9.7.2 Transfer Functions: Mathematically Describing Filters in **ECE Confidential Cracking The Code** Understand following terms

- *Transfer function*: The magnitude of the transfer function $|H(j\omega)|$ describes how the amplitude of the output signal varies with frequency relative to the input signal, while the phase $\angle H(j\omega)$ describes the phase shift between input and output signals.
- *Low-pass filter transfer function*:

$$H_{LP}(j\omega) = \frac{1}{1 + j\omega\tau} = \frac{1}{1 + j\omega RC}$$

  for an RC filter.
- *High-pass filter transfer function*:

$$H_{HP}(j\omega) = \frac{j\omega\tau}{1 + j\omega\tau} = \frac{j\omega RC}{1 + j\omega RC}$$

  for an RC filter.
- *Cutoff frequency*: The cutoff frequency occurs where the magnitude of the transfer function is $\frac{1}{\sqrt{2}} \approx 0.707$ of its maximum value, which corresponds to -3 dB. For an RC filter $\omega_c = 2\pi f_c = \frac{1}{RC}$.

Prelab Deliverable #1a

Explain what low pass and high pass filters are and how they are used in electronic systems. Describe at least three practical applications of each type of filter.

Prelab Deliverable #1b

For the following filter circuits, identify which is a low pass filter and which is a high pass filter. Explain your answer in your own words (Hint: consider outputs with very low frequency, such as DC, and very high frequencies).

(a) Circuit 1 (b) Circuit 2

Figure 1: RC Filter Circuits

Prelab Deliverable #1c

Calculate the cutoff frequency f_cFor an RC low pass filter with $R = 1$ kΩ and $C = 0.1$ μF .

Prelab Deliverable #1d.A

For an RC low pass filter with $R = 1$ kΩ and $C = 0.1$ μF, calculate the theoretical magnitude of the transfer function in dB and phase shift (degrees) at $f_c/10$.

Prelab Deliverable #1d.B

For an RC low pass filter with $R = 1$ kΩ and $C = 0.1$ μF, calculate the theoretical magnitude of the transfer function in dB and phase shift(degrees) at f_c.

Prelab Deliverable #1d.C

For an RC low pass filter with $R = 1$ kΩ and $C = 0.1$ μF calculate the theoretical magnitude of the transfer function in dB and phase shift (degrees) at $10 \times f_c$.

Prelab Deliverable #1e

Complete and modify the provided MATLAB starter code to create Bode plots (magnitude and phase) for a low pass RC filter with $f_c = 1$ kHz, and double check your answers 1d. Create one for magnitude and a separate one for phase. Insert the two plots in pre-lab report.

The starter MATLAB code for Bode plot. Complete the line of code indicated by => and plot the high pass filter response in a similar format. Include your plots in your lab report:

```
% Bode Plot
% RC Filter Bode Plot Generator
% Parameters
R = 1000;    % 1kOhm resistor
% =>C = ...... % COMPLETE THIS LINE OF CODE  Calculate the capacitance for ~1kHz cutoff
f = logspace(1, 5, 1000); % Frequency range from 10Hz to 100kHz
w = 2*pi*f;  % Angular frequency

% Low pass filter transfer function
H_lp = 1./(1 + 1j*w*R*C); %

% High pass filter transfer function
% =>H_hp = ........ % COMPLETE THIS LINE OF CODE

% Magnitude in dB
mag_lp_db = 20*log10(abs(H_lp));
mag_hp_db = 20*log10(abs(H_hp));

% Phase in degrees
phase_lp = angle(H_lp)*180/pi;
phase_hp = angle(H_hp)*180/pi;

% Plotting
figure(1);
semilogx(f, mag_lp_db);
title('Low Pass Filter - Magnitude Response');
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
grid on;
% Add a horizontal line at -3 dB
hold on;
semilogx([min(f), max(f)], [-3, -3], 'r--');
hold off;

figure(2);
semilogx(f, phase_lp);
title('Low Pass Filter - Phase Response');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
```

```
grid on;

figure(3);
%=>% Complete the plotting for high pass filter
%=>% [Add your code here]
```

Code can be found here: **Bode Plot Tarter Code**.

2 Lab Procedure: Low Pass Filter

IMPORTANT

This lab must be completed during the scheduled lab session. While collaboration and discussion with classmates is encouraged, each student must complete and submit their own work. If you encounter difficulties, first consult with your peers for assistance. If questions remain unresolved, please seek help from the student assistant or TA, who are available to provide guidance throughout the lab period.

2.1 Filter Circuit Construction

1. Measure the values of R and C.
2. Construct the low pass RC filter using a 1 kΩ resistor and a 0.1 μF capacitor as shown in Figure 2.

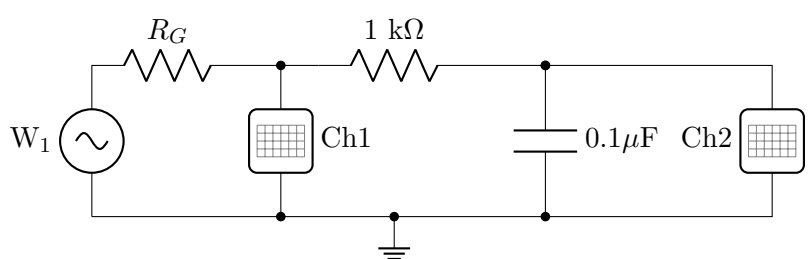

Figure 2: Connection diagram for low pass filter measurement

Lab Deliverable #1a

Take a clear photograph of your low pass filter measurement circuit setup **along with a piece of paper on which the measured values of R and C are shown, including the measurement errors**.

2.2 Low Pass Filter Time Domain Measurement

1. Configure the signal generator to produce a 100 Hz sine wave with 2V peak-to-peak amplitude.
2. Observe both the input (CH1) and output (CH2) signals on the oscilloscope.

3. Measure the peak-to-peak amplitude of both signals and calculate the magnitude of the transfer function:

$$|H(j\omega)| = \frac{V_{out}}{V_{in}} \tag{1}$$

where ω is the frequency produced by the signal generator.

4. Convert this magnitude to decibels (dB):

$$|H(j\omega)|_{dB} = 20\log_{10}\left(\frac{V_{out}}{V_{in}}\right) \tag{2}$$

5. Measure the phase difference between the input and output signals. You can use the cursors to measure the time delay (Δt) between corresponding points on the input and output waveforms, then convert to phase angle:

$$\phi = \frac{\Delta t}{T} \times 360 \tag{3}$$

where T is the period of the waveform.

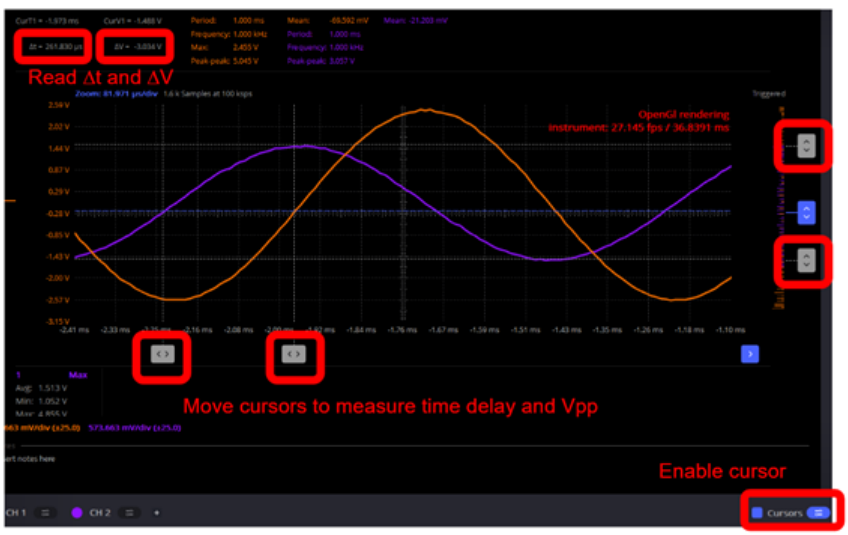

Figure 3: Steps to measure time delay and peak-to-peak amplitude using cursors

6. Record screenshots of the oscilloscope display showing both input and output signals **and clearly mark the amplitude and phase measurements on each screenshot**.

7. Repeat steps 1-6 with sine waves at 500 Hz, 1 kHz, 2 kHz, 5 kHz, and 10 kHz.

Lab Deliverable #1b

Insert measured values in Table 1 with your measurements.

Lab Deliverable #1c.A

Lowpass at 100 Hz: Provide your screenshots of the oscilloscope display showing both input and output signals.Include markups showing amplitude and phase measurements.

Table 1: Low Pass Filter Measurements

Frequency (Hz)	V_{in} **(V)**	V_{out} **(V)**	$\lvert H(j\omega)\rvert$	$\lvert H(j\omega)\rvert_{dB}$	Phase (°)
100					
500					
1000					
2000					
5000					
10000					

Lab Deliverable #1c.B

Lowpass at 500 Hz: Provide your screenshots of the oscilloscope display showing both input and output signals. Include markups showing amplitude and phase measurements.

Lab Deliverable #1c.C

Lowpass at 1 kHz: Provide your screenshots of the oscilloscope display showing both input and output signals. Include markups showing amplitude and phase measurements.

Lab Deliverable #1c.D

Lowpass at 2 kHz: Provide your screenshots of the oscilloscope display showing both input and output signals. Include markups showing amplitude and phase measurements.

Lab Deliverable #1c.E

Lowpass at 5 kHz: Provide your screenshots of the oscilloscope display showing both input and output signals. Include markups showing amplitude and phase measurements.

Lab Deliverable #1c.F

Lowpass at 10 kHz: Provide your screenshots of the oscilloscope display showing both input and output signals. Include markups showing amplitude and phase measurements.

2.3 Low Pass Filter Frequency Domain Measurement

The Network Analyzer in Scopy provides an automated way to measure the frequency response of filters. It sweeps a sine wave through a range of frequencies and measures the magnitude and phase of the transfer function at each frequency.

1. Navigate to the Network Analyzer tab in Scopy.
2. Configure the Network Analyzer with the following settings:
   - Frequency range: 10 Hz to 100 kHz (logarithmic sweep)
   - Number of points: 1000
   - Amplitude: 2V peak-to-peak
   - Channel 1: Set as reference (input)
   - Channel 2: Set as measurement (output)
   - Display: Set to show both magnitude (in dB) and phase. Set the minimum phase at -90 the maximum phase at 90. Magnitude axis can be set from -50 dB to 10dB.
3. Click "Run" to start the measurement.

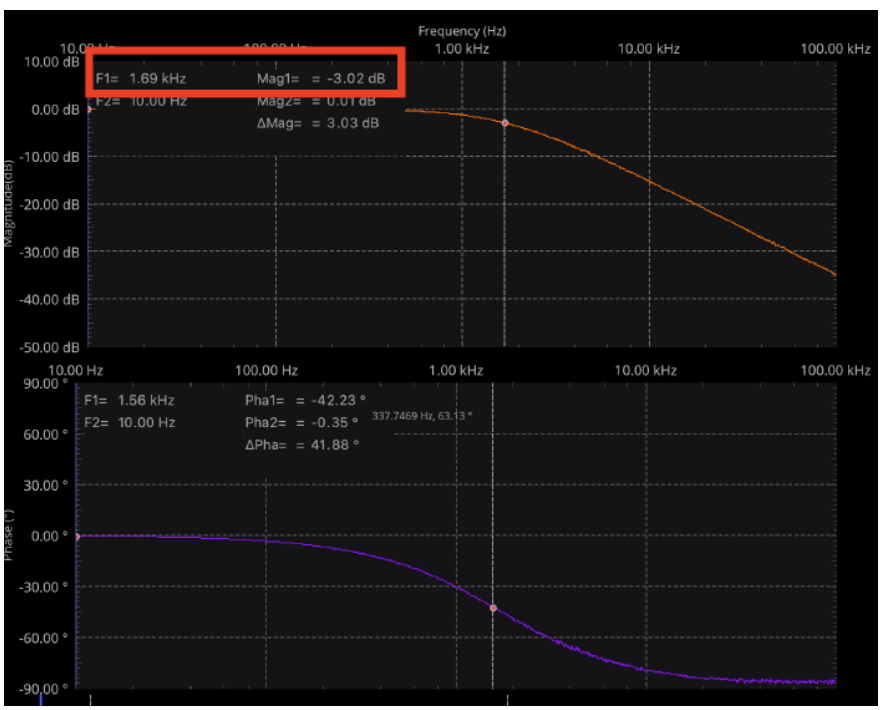

Figure 4: Example Network Analyzer display for lowpass filter

4. Export the frequency response data to a CSV file named "LPFrequencyResponse.csv".

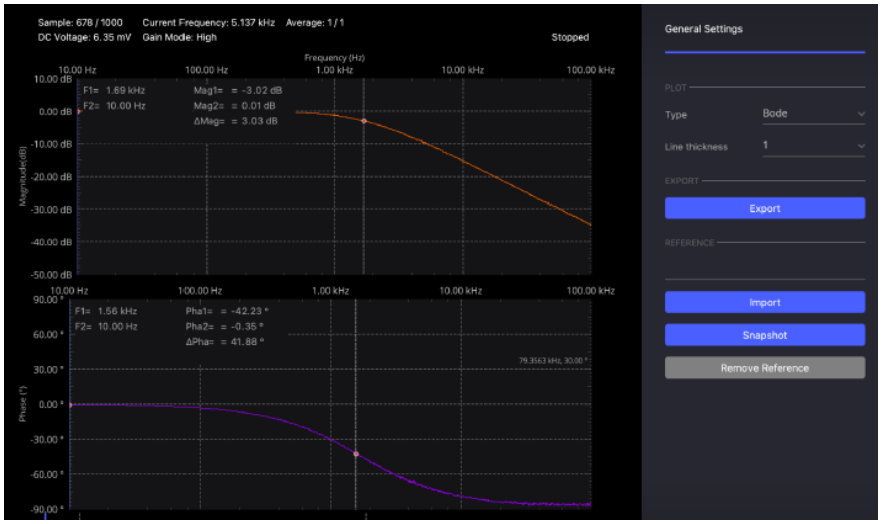

Figure 5: Example Network Analyzer export display

	A	B	C	D
	;Scopy version	dda5ec6		
	;Exported on	Thursday April 24/04/2025		
	;Device	M2K		
	;Nr of samples	1000		
	;Sample rate	0		
	;Tool	Network Analyzer		
	;Additional Inforr	Reference channel: 1		
	Sample	Frequency(Hz)	Magnitude(dB)	Phase(°)
	0	10	0.013275	-0.345637
	1	10.0926	0.0158275	-0.369388
	2	10.1861	0.0137163	-0.33007
	3	10.2804	0.0130038	-0.346386
	4	10.3757	0.014436	-0.368243
	5	10.4718	0.0123008	-0.385
	6	10.5688	0.0135095	-0.389536
	7	10.6666	0.0145323	-0.344738

Figure 6: Example Network Analyzer export file

Lab Deliverable #1d

Take a screenshot of your Network Analyzer setup in Scopy and include it in your lab report.

3 Lab Procedure: High Pass Filter

3.1 Circuit Construction

Construct the highpass RC filter using a 1 kΩ resistor and a 0.1 μF capacitor as shown in Figure 7.

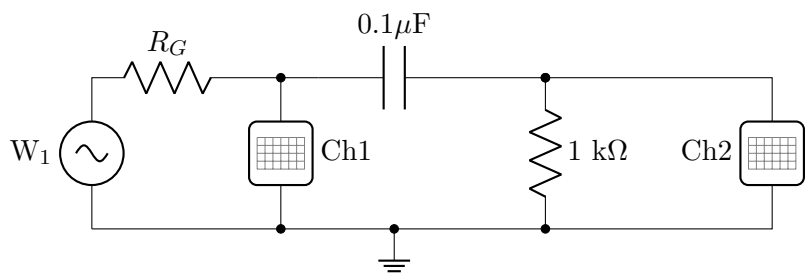

Figure 7: Connection diagram for high pass filter measurement

Lab Deliverable #2a

Take a clear photograph of your high pass filter measurement circuit setup and include it in your lab report.

3.2 High Pass Filter Time Domain Measurement

1. Configure the signal generator to produce a 100 Hz sine wave with 2V peak-to-peak amplitude.
2. Observe both the input (CH1) and output (CH2) signals on the oscilloscope.
3. Measure the peak-to-peak amplitude of both signals and calculate the magnitude of the transfer function:
$$|H(j\omega)| = \frac{V_{out}}{V_{in}} \tag{4}$$
4. Convert this magnitude to decibels (dB):
$$|H(j\omega)|_{dB} = 20\log_{10}\left(\frac{V_{out}}{V_{in}}\right) \tag{5}$$
5. Measure the phase difference between the input and output signals. You can use the cursors to measure the time delay (Δt) between corresponding points on the input and output waveforms, then convert to phase angle:
$$\phi = \frac{\Delta t}{T} \times 360 \tag{6}$$
where T is the period of the waveform.
6. Record screenshots of the oscilloscope display showing both input and output signals **and clearly mark the amplitude and phase measurements on each screenshot**.
7. Repeat steps 1-6 with sine waves at 500 Hz, 1 kHz , 2 kHz, 5 kHz, and 10 kHz.

Lab Deliverable #2b

Insert measured values in Table 2 with your measurements.

Table 2: High Pass Filter Measurements

Frequency (Hz)	V_{in} **(V)**	V_{out} **(V)**	$\lvert H(j\omega)\rvert$	$\lvert H(j\omega)\rvert_{dB}$	Phase (°)
100					
500					
1000					
2000					
5000					
10000					

Lab Deliverable #2c.A

Highpass at 100 Hz: Provide your screenshots of the oscilloscope display showing both input and output signals. Include markups showing amplitude and phase measurements.

Lab Deliverable #2c.B

Highpass at 500 Hz: Provide your screenshots of the oscilloscope display showing both input and output signals. Include markups showing amplitude and phase measurements.

Lab Deliverable #2c.C

Highpass at 1 kHz: Provide your screenshots of the oscilloscope display showing both input and output signals. Include markups showing amplitude and phase measurements.

Lab Deliverable #2c.D

Highpass at 2 kHz: Provide your screenshots of the oscilloscope display showing both input and output signals. Include markups showing amplitude and phase measurements.

Lab Deliverable #2c.E

Highpass at 5 kHz: Provide your screenshots of the oscilloscope display showing both input and output signals. Include markups showing amplitude and phase measurements.

Lab Deliverable #2c.F

Highpass at 10 kHz: Provide your screenshots of the oscilloscope display showing both input and output signals. Include markups showing amplitude and phase measurements.

3.3 High Pass Filter Frequency Response Measurement

1. Repeat the Network Analyzer measurements for the high pass filter.
2. Again, identify the -3 dB point and measure the phase at the cutoff frequency. For a first-order high pass filter, the phase shift at the cutoff frequency should be approximately +45 degrees.
3. Export the frequency response data to a CSV file named "HPFrequencyResponse.csv".

Lab Deliverable #2d

Capture screenshots of both the magnitude and phase plots for the high pass filter. Annotate the plots to indicate the cutoff frequency and the phase at that frequency. Discuss how your measured cutoff frequency compares with the theoretical value calculated in the prelab.

4 Post-Lab Analysis and Report

4.1 MATLAB Analysis of Filter Frequency Response

Using MATLAB, compare your theoretical and measured filter responses.

Write a MATLAB script that:

1. Load the frequency response data, measured with the Network Analyzer, for both filters
2. From the Network Analyzer measurements, create Bode plot (magnitude and phase) for both filters
3. On the same plot, using the measured component values, display the calculated Bode plot (magnitude and phase) for both filters
4. On this plot, display the discrete values taken in from time domain measurements reported in Tables 2 and 1.
5. Includes clear legends, titles, and axis labels for all plots

Lab Deliverable #3a.A

For the low pass filter, insert composite Bode plot (magnitude and phase) comparing theoretical and measured responses

Lab Deliverable #3a.B

For the high pass filter, insert composite Bode plot (magnitude and phase) comparing theoretical and measured responses

Lab Deliverable #3b

Calculate the percentage error in the cutoff frequency for both filters and discuss possible sources of error, including component tolerances, parasitic elements, and measurement limitations.

4.2 Understanding Filter Characteristics

Lab Deliverable #3c

In your own words,for both low pass and high pass filters:

1. Explain how the magnitude response changes as the frequency increases, with reference to your measured data
2. Explain how the phase response changes across the frequency range and why this is important in signal processing applications
3. Describe the filter's behavior in terms of asymptotic slopes (dB/decade) at frequencies far above and far below the cutoff frequency
4. Discuss the significance of the dB scale and logarithmic frequency representation in analyzing filter performance

5 Submission Instructions

Submit your complete lab report via the specified platform by the due date. Your report should include all deliverables described above.

6 Grading Criteria

Your lab will be graded according to the following criteria:

- Completeness (all deliverables included) - 50%
- Analysis and interpretation quality - 25%
- Correctness of measurements and calculations - 15%
- Quality of documentation (photos, screenshots, graphs) - 10%